Engine Model(s): PT6A

72-50-03

PT INTERSTAGE STATOR - MAINTENANCE PRACTICES

1. General

A. Maintenance personnel should make reference to the INTRODUCTION section and Chapter 70-00-00, STANDARD PRACTICES of this manual to familiarize themselves with general procedures.

B. Install suitable protective caps/covers over all disconnected tubes/lines and component openings.

C. Lockwire shall comply with specification AMS 5687, heat and corrosion resistant steel wire MS9226-03, which unless otherwise specified is 0.025 inch diameter, and will not be specified in instructions.

2. Consumable Materials

The consumable materials listed below are used in the following procedures.

Item No.	Name
PWC05-110	Lockwire
PWC06-015	Beeswax

3. Special Tools

The special tools listed below are used in the following procedures.

Tool No.	Name
PWC32212	Pin, Alignment

4. Fixtures, Equipment and Supplier Tools

Not Applicable

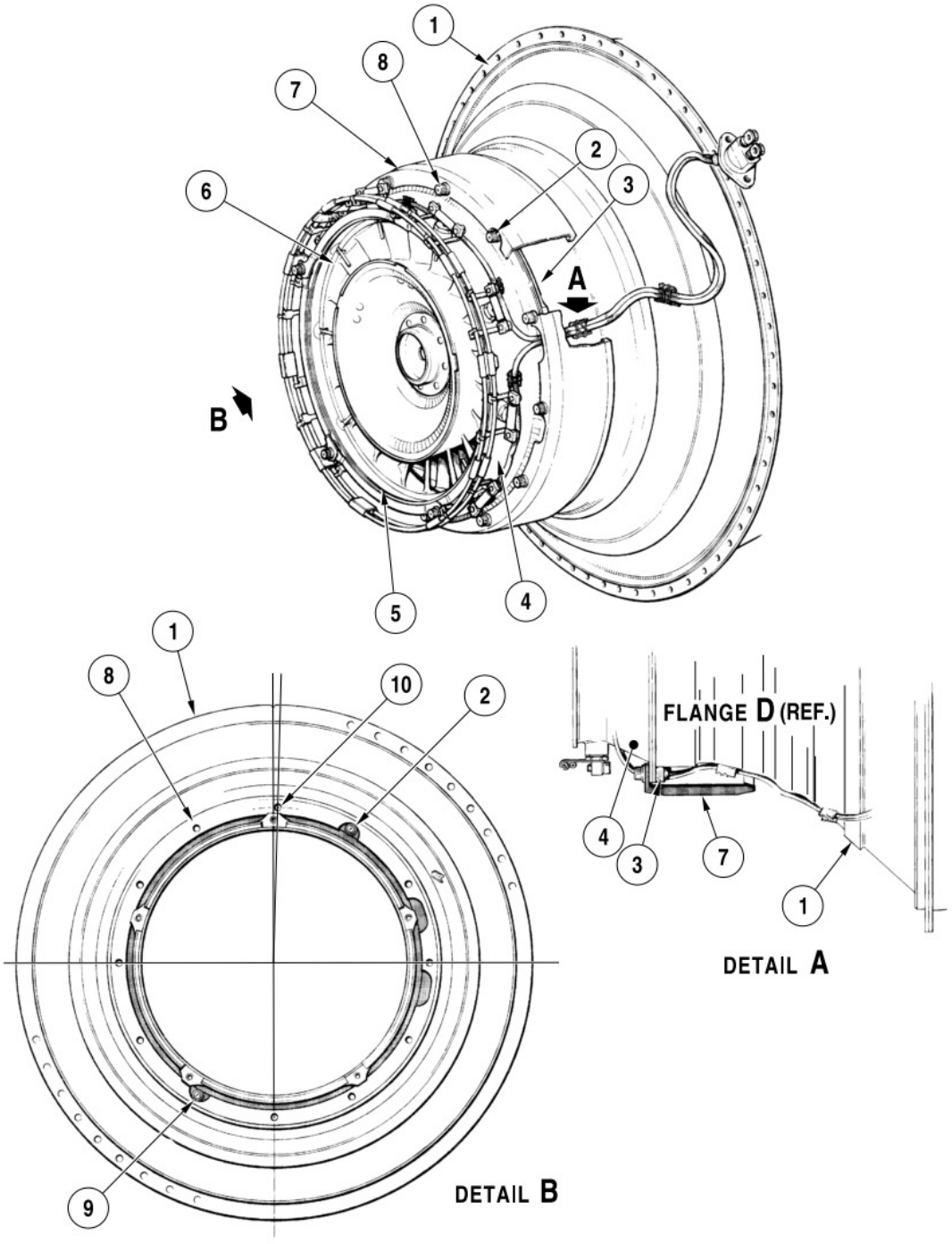
5. Removal/Installation (Heavy Maintenance)

NOTE: Disassembly of the power section should only be undertaken whenever hot section distress requiring corrective action is indicated and/or whenever the performance requirements necessitate power turbine vane ring removal. Whenever a replacement power section is installed on an engine at maintenance, the original power turbine vane ring and interstage baffle assembly should be retained and installed on the new power section if practicable. This requirement is to make sure the throat area matches between the compressor turbine and power turbine vane rings which can affect the operating parameters of the engine.

A. Removal Post-SB1220 Power Turbine Stator (Ref. Fig. 201)

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Figure 201 Removal/Installation of the Power Turbine Stator Assembly



Post-SB1220

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1. Exhaust duct
2. Bolt
3. Half-rings
4. Stator Housing
5. Interstage Sealing Ring(s)
6. Vane Ring Assembly
7. Containment Ring
8. Bolt
9. Bolt
10. Off-set Hole

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(1) Remove power section and install in engine stand (Ref. 72-00-00).

(2) Cut lockwire and remove T5 thermocouple harness, bus-bar assembly, T5 thermocouple probes, containment ring, seal retaining plates (Ref. 77-20-01) and interstage sealing ring(s) (5).

NOTE: For PT6A-21 engines and Post-SB1177 PT6A-27 and PT6A-28 engines: the bus-bar assembly, interstage sealing ring(s) need not be removed unless they are to be replaced. If Pre-SB1218 interstage sealing rings are to be removed, mark position of the sealing ring gaps on the power turbine housing with an approved marker (Ref. 70-00-00, MAINTENANCE PRACTICES, Marking of Parts).

(3) Remove two bolts (2 and 9), at 1 o'clock and 7 o'clock positions, retaining power turbine stator housing (4) to exhaust duct (1) at Flange D. Remove two half rings (3).

(4) Withdraw assembly and vane ring and interstage baffle assembly (6) from power section by pulling on vanes.

(5) Place assembly on bench with leading edge of vanes facing upward. Lift stator housing (4) from vane ring assembly (6).

B. Installation of Power Turbine Stator (Ref. Fig. 201)

CAUTION: ON ALL ENGINE CONFIGURATIONS, EXERCISE EXTREME CARE NOT TO DAMAGE THE T5 PROBES WHEN FITTING THE POWER TURBINE VANE RING. ON ENGINES INCORPORATING THE INTENT OF SB1212 (EIGHT T5 PROBES ONLY), MAKE SURE THAT NONE OF THE PROBES ARE ALIGNED WITH EITHER OF THE TWO CLOSELY SPACED SLOTS IN THE POWER TURBINE VANE RING.

(1) Locate power turbine vane ring (6) in stator housing (4). Align slots in vane ring with respective thermocouple probe holes in stator housing and mate lugs in housing with slots in vane ring. Parts should mate freely and without force.

(2) Holding vanes of vane ring firmly to prevent separation of ring and housing, install complete assembly on Flange D of exhaust duct (1). Use care to prevent chafing T5 thermocouple harness.

(3) Locate two half rings (3) at front face of Flange D and temporarily secure vane ring (6), stator housing (4) and containment ring (7) to exhaust duct (1) and ring halves with alignment pins (PWC32212).

(4) Secure vane ring (6) and stator housing (4) to Flange D with single bolt at 1 o'clock and 7 o'clock positions.

(5) Remove alignment pins (PWC32212). Secure containment ring (7) together with vane ring and housing to Flange D and ring halves with 10 bolts (8). Tighten bolts and torque 32 to 36 lb.in., and lockwire (PWC05-110).

(6) Install T5 thermocouple probes (if removed), bus-bar assembly and the seal ring retaining plates (Ref. 77-20-01)

(7) Install the interstage sealing ring(s)(5). Make sure the interstage sealing ring(s) are centralized in the stator housing.

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NOTE: 1. For Pre-SB1218 sealing rings, make sure the ring gaps are diametrically opposite as noted at disassembly. For Post-SB1218 sealing rings, make sure the side of the ring marked "PRESS" faces the rear of the engine.

NOTE: 2. To keep the sealing ring centralized, insert suitable sized pieces of beeswax (PWC06-015) in the cavity of the housing, at the 6 o'clock position, to support the ring.

(8) Connect T5 harness to bus-bar, and lockwire harness to exhaust duct (Ref. 77-20-01).

(9) Do a functional check on the T5 system (Ref. 77-20-01).

(10) Install the power section (Ref. 72-00-00).

6. Inspection/Check

A. General

Acceptance criteria outlined herein is based on structural integrity considerations and apply only if engine performance /handling was satisfactory prior to HSI. Replace stator assembly if its condition appears to be the cause for loss of performance.

Stator assemblies considered unacceptable for continued service may be sent to PWC for possible repair.

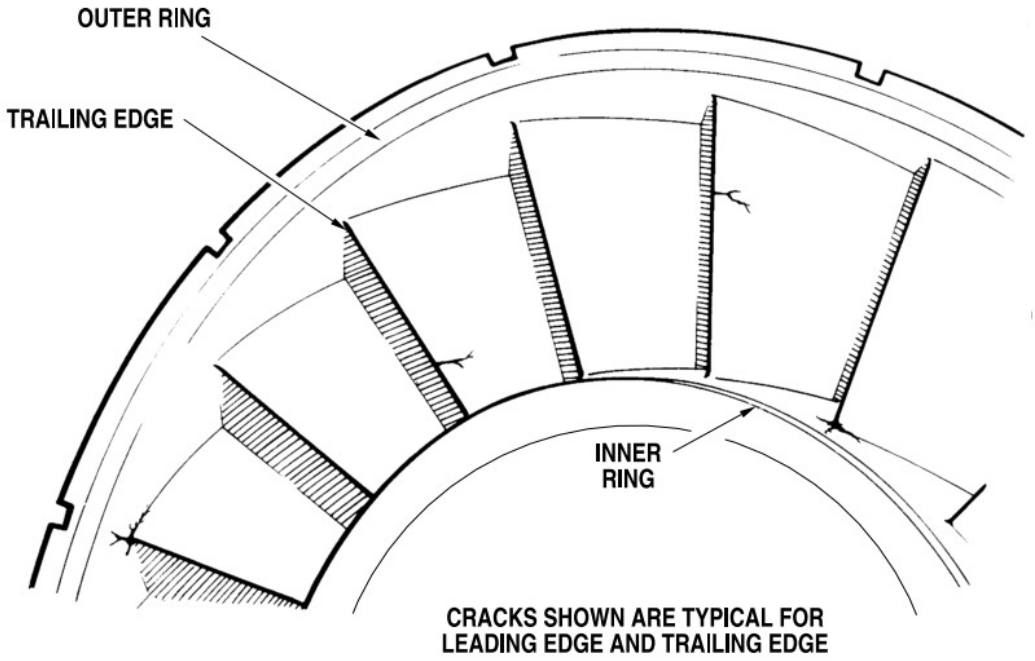
Pratt & Whitney Canada Corp.
7007 de la Savanne
St. Hubert , Quebec
Canada
J3Y 3X7

Attention: Receiving Area 5-1

B. Power Turbine Stator Assembly (Ref. Figs. 202 and 203)

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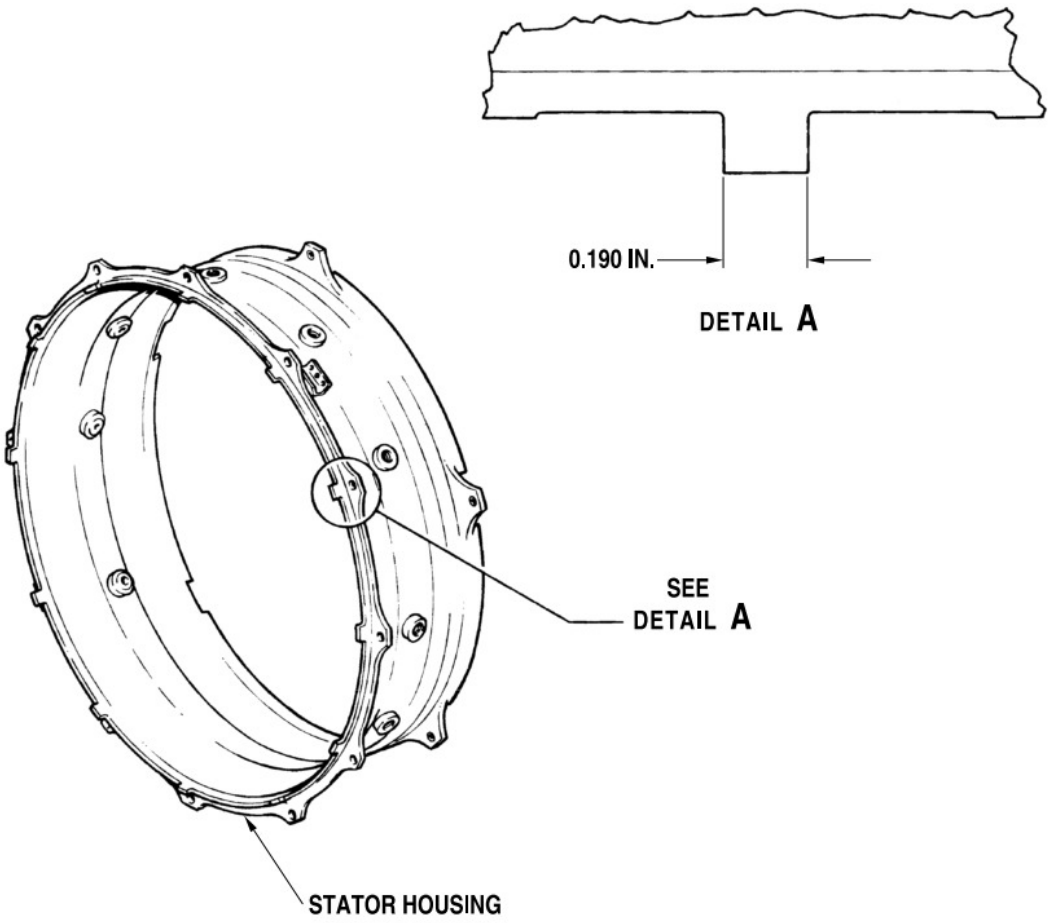
Figure 202 Power Turbine Vane Ring Inspection (Typical)



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Figure 203 Power Turbine Stator Housing Lug Wear Limits



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(1) Inspect the power turbine stator assembly (Ref. Table 201).

Table 201 Power Turbine Stator Assembly Inspection

Item	Max. Serviceable	Action
VANE RING		
Cracks:		
Leading and/or trailing edges	Cracks up to 0.400 in. long on any number of vanes.	Replace
Outer ring	0.600 in. long	Replace
Inner ring	One crack through entire length is acceptable provided crack is not open more than 0.020 in.	Replace
Erosion or Pitting:		
Airfoil Surfaces	Erosion or pitting, 0.250 sq.in. by 0.010 in. deep on any airfoil surface, any number of vanes.	Replace
POWER TURBINE STATOR HOUSING		
Fretting wear:		
Lugs	Check with vane ring for circumferential movement. Average 0.190 inch minimum lug width acceptable.	Repairable
Sealing Face	Acceptable providing face shows witness of contact around complete circumference when checked against sealing ring.	Replace
NOTE: Light stoning to remove localized high spots on sealing face (i.e. witness mark from sealing ring joint) is acceptable.		
INTERSTAGE BAFFLE	Examine for correct attachment. Replace loose rivets.	Repairable
INTERSTAGE SEALING RINGS		
Fretting		

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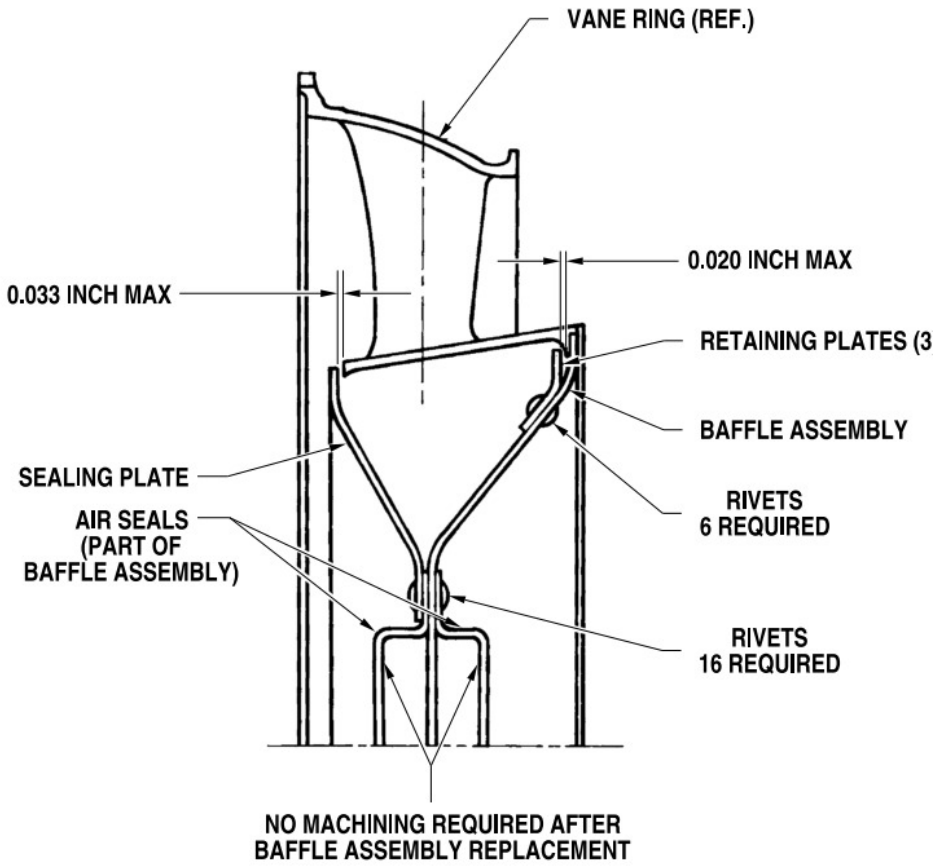
Item Wear:	**Max. Serviceable**	**Action**
Inside Diameter	Acceptable provided end faces of ring do not meet when ring is installed in sealing location over CT shroud housing.	Replace
Sealing Face	Acceptable provided face shows witness of contact around complete circumference when checked against true flat surface. Install original or new ring(s) individually in correct location and orientation on compressor turbine shroud housing. Check gap between ring and housing sealing diameter; maximum acceptable gap in free state is 0.002 inch over not more than 120 degrees of arc, remaining area to be light tight. Reject ring and/or housing if this limit is exceeded.	Replace
NOTE: Light stoning of ring ID and compressor turbine shroud housing sealing surface to remove localized high spots is acceptable.		

7. Approved Repairs

A. Replacement of Interstage Baffle Rivets (Ref. Fig. 204)

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Figure 204 Replacement of Interstage Baffle Rivets



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(1) Remove the 16 rivet heads at the baffle assembly by grinding or drilling, and carefully punch out the rivet stems with a parallel pin punch.

(2) Remove the sealing plate and the two interstage air seals from the baffle on the vane ring.

(3) Remove the six rivet heads and carefully press out the rivet stems with a parallel pin punch. Remove the baffle assembly and retaining plates from the vane ring.

(4) Install the baffle on the vane ring with three retaining plates and six rivets. Maintain the dimensions indicated.

(5) Locate the two interstage air seals on the baffle, and position the sealing plate relative to the dimension indicated, relative to the vane ring.

(6) Fasten the baffle assembly on the vane ring with 16 rivets.

NOTE: Use care when riveting to avoid distortion of the sealing plate or the baffle assembly.

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